

# Response to reviewers

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Thank you for the detailed reports. The revision restricts the identifiability claim, separates the observation model from the communication model, and narrows the supporting simulations. The regional dictionary is now an operational encoding map, not a synonym for preferred phase. A short finite-memory pathway interaction replaces the disconnected idea-network development. All locations below refer to the compiled revised manuscript supplied with this letter; the page references are generated from its final labels. The theory has no human or animal validation, and no universal mechanistic uniqueness or robust neural parameter recovery is claimed.

## Reviewer 1: major comments

### 1. Appendix B and Test 6 are loosely connected and too long.

The exploratory idea-network/creativity program and its associated test are omitted. Neither the main abstract nor the Discussion makes an idea-generation claim. Scripts remain separate reproducibility files rather than long code listings in the article. Marker, ordering, and serial-path checks are in the supplement. Dictionary maps remain because they define the content carried by the transfer operator (Section ??, p. ??). The short pathway interaction is directly tied to that operator and has a derived factorial contrast (Section ??, p. ??); it is not a theory of experience. Its flexible nonlinear comparator makes the limits of discrimination explicit (Section ??, p. ??).

### 2. Provide code and make it publicly available.

*Implemented part:* the complete source archive includes executable simulation, verification, numerical-macro generation, dependency requirements, all figures, per-realization results, a checksummed manifest, and a release checklist. All figures were regenerated and the verification script passed. See Transparency (p. ??) and Supplement Section 1.

*Outstanding part:* a public archival deposit has not been made. The present working environment provides no configured public research-repository publication route. The code-availability statement explicitly says the deposit is pending; the review archive is not described as publicly archived. This reviewer requirement is therefore only partially fulfilled and must be completed before claiming full compliance. The release checklist gives the exact files and statement to update after a real permanent URL is obtained.

### 3. Response letter and manuscript disagree about uniqueness/recovery.

This letter contains no claim that the integration uniquely distinguishes itself from other theories, or recovers parameters accurately from neural recordings. Exact-feature coupling recovery is conditional, observed interval coverage is reported even when below nominal, and the flexible model predicts comparably (Section ??, p. ??). Source-mixing recovery is biased (Section ??, p. ??). The full-law identifiability condition is explicit (Section ??, p. ??). All changes are located by section and compiled page.

## Reviewer 1: minor comments

1. **Title/header.** The revised title specifies the single-band limit and separate identification. The short-title metadata and supplementary running header use “Single-band delay–preference non-identifiability.” The submission class uses the journal’s generic under-review header, containing no old title (title page; Section ??, p. ??). The abstract now states the ridge first and says the access–trace split is built into the learning rule.
2. **Terminology.** The operational term is consistently “unreported processing.” The motivation involving subconscious/conscious communication is explicitly translated into unreported and report-associated pathways without assigning experiential status (Section ??, p. ??).

3. **Reviewer-facing sentences.** Reviewer-specific discussion is confined to this letter. The Transparency section states availability for readers without describing it as satisfying a reviewer request (Transparency (p. ??)).
4. **Imposed dissociation.** The abstract explicitly says that the access–trace split is imposed by the learning rule. The main explanation is at Section ??, p. ??; Supplement Section 5 computes the trace from five actual updates instead of inserting a fixed demonstration value.

## Reviewer 2: major comments

### 1. Restrict title/headline to the theorem’s domain.

The title, abstract, scope paragraph, proposition, and Discussion restrict the result to a stationary single-frequency likelihood that depends on preference and conduction delay through their effective combination (title page; Section ??, p. ??; Section ??, p. ??; Section ??, p. ??). The text does not claim general blindness of all phase-based connectivity. A new comparison table states what would discriminate this account from CTC, access, recurrent-processing, and coherence-as-consequence models without claiming that success adjudicates those theories (Section ??, p. ??). The supplementary marker/ordering material is outside the negative theorem’s domain.

### 2. Separate parameter non-identifiability from causal/mechanistic identification.

Five questions are defined separately at Section ??, p. ??. The phase–report association conclusion now concerns nonunique inference against specified alternative explanations, and is explicitly not derived from the rank-one result alone (Section ??, p. ??). Raw observed phase is distinguished from modeled arrival error; an unchanged gate parameterization alone does not prove that a measured lag estimates the effective offset.

### 3. Clarify the gain from “gauge orbit.”

The transformation is presented as a standard parameter ridge. The gauge name denotes its equivalence class only and adds no theorem beyond reparameterization (Section ??, p. ??). It is not described as a local physical gauge symmetry.

### 4. Remove “the only design.”

Arrival-phase intervention is one gate-testing strategy. Independent latency, constrained multi-frequency observations, and a known chirp can add transverse information under their explicit assumptions (Section ??, p. ??). A receiver-only reset changes a known additive phase and does not separately identify delay/preference. Neither waveform shape nor harmonics automatically adds rank: the shared-offset and proportional-harmonic cases are contrasted.

### 5. Classify simulations; move secondary material.

Table ?? (p. ??) distinguishes consistency, conditional parameter recovery, robustness, restricted generator comparison, unclaimed general model recovery, and proposed empirical analyses. Four secondary figures and analytic details are in the supplement. The main article contains three figures for delayed transfer, coupling recovery, and practical recovery. The four-population result is not represented as independent biological validation; the retained minimal trace split is identified as imposed (Section ??, p. ??).

### 6. Prominence of practical recovery limitations.

Structural and practical identifiability are separated early (Section ??, p. ??) and in a dedicated primary analysis (Section ??, p. ??), with conclusions in Section ??, p. ??. Matched ablations isolate source mixing and common drive. Importantly, a 16-Hz harmonic is removed by the specified ideal 6–10 Hz filter: the revised manuscript no longer attributes this simulation’s distortion to that harmonic. All predictor comparisons now use held-out trials, and a Fourier comparator replaces reliance on a constant alone. Concentration remains biased under mixing/common drive; recovered offset alone does not validate the recording model.

**7. Strengthen anti-circularity and access-circuit stability.**

Section ??, p. ?? prescribes four independent stages: localization in a separate session/dataset, decoder/pathway/kernel calibration, parameter fitting, and confirmatory testing. Stability across tasks, sessions, and participants is prespecified using calibration only. Failure invokes an inconclusive declaration or an already registered participant-specific rule, without outcome-guided re-localization. Rejection rules are now in a table: no sensitivity effect after a verified arrival-phase move, transfer on a structurally closed path, timing rescue of a null content direction, trace change with the learning flag closed, and reading  $\eta$  from an uncontrolled mixture (Section ??, p. ??).

**8. Separate implemented and proposed statistical models.**

The implemented Gaussian sensitivity derivative has an explicit class-mean convention (Section ??, p. ??). The trial sensitivity term is a latent choice-likelihood parameter, not a measured per-trial score. Behavioral sensitivity/criterion regression and joint decoder/confidence/error likelihood remain proposed. Coupling recovery and delayed readout fitting are implemented with known features/maps/kernels (Section ??, p. ??; Section ??, p. ??). No completed joint awareness model is claimed.

**9. Synthetic power is not empirical sample-size justification.**

The earlier illustrative and participant-level power analyses are removed from this focused revision. Section ??, p. ?? lists the quantities needed for a future empirical calculation and makes no sample-size recommendation. Counts in the simulation methods are generation settings, not recruitment guidance. Reported interval coverage is a finite synthetic operating characteristic, not an empirical power calculation.

**10. Narrow the manuscript and remove the distant idea-network program.**

The idea-network/creativity program is omitted entirely. The dictionary material is limited to operational encoding, translation, conduction/processing, and a short cross-path finite-memory extension (Section ??, p. ??; Section ??, p. ??). This extension is needed only to test communication beyond separable pathway contributions; it is not necessary for the single-band theorem and is not derived from it. Its factorial restriction and flexible comparator preserve a direct link to the central transfer model. Secondary latency/ordering demonstrations are confined to the supplement.

## **Reviewer 2: minor comments**

- 1. Terminology and notation.** Table ?? (p. ??) separates observed phase, arrival error, effective offset, timing, content maps, boundary-route score, and trace. The former overlapping symbols are reduced rather than retained as undefined aliases. Section ??, p. ?? distinguishes consistency from receptive alignment and gives the actual cross-spectral wPLI definition with magnitude weights.
- 2. Temporal dependence/PPC.** The dense-sampling caveat is in the early assumptions subsection (Section ??, p. ??), not buried in the numerical section.
- 3. Conduction versus signal-processing phase delay.** These are distinguished in Section ??, p. ?? and Section ??, p. ??; distributed processing adds its own transfer phase and cannot simply be relabeled conduction.
- 4. Illustrative thresholds.** All thresholds require independent pilot calibration (Section ??, p. ??; Section ??, p. ??). A sufficiently precise equivalence/meaningful-effect bound, rather than failure to reject a null, governs rejection of the registered effect.

5. **Shortening/consolidation.** Global assumptions are consolidated at Section ??, p. ??. The main manuscript removes long code listings, the idea-network program, secondary ordering/marker sections, and synthetic participant-power analyses. The supplement contains secondary methods rather than an additional consciousness theory.